



K-BAC 2050

Printed: 12/07/2017
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SAFETY DATA SHEET

1. PRODUCT AND COMPANY IDENTIFICATION

Product Code: 211516
Product Name: K-BAC 2050
Category: Biocide
Company Name: Water Science Technologies, LLC
1701 Vanderbilt Road
Birmingham, AL 35234
Phone Number: (866)284-9244
Emergency Contact: ChemTel, Inc. (US) (800)255-3924
ChemTel, Inc. (Int'l) (813)248-0585
Information: Company EPA #: 88714

2. HAZARDS IDENTIFICATION

Acute Toxicity: Inhalation, Category 2
Acute Toxicity: Oral, Category 3
Serious Eye Damage/Eye Irritation, Category 1
Skin Corrosion/Irritation, Category 1B
Respiratory Sensitization, Category 1
Skin Sensitization, Category 1
Specific Target Organ Toxicity (single exposure), Category 3
Aquatic Toxicity (Acute), Category 1
Aquatic Toxicity (Chronic), Category 2



GHS Signal Word: Danger

GHS Hazard Phrases: H301 - Toxic if swallowed.
H314 - Causes severe skin burns and eye damage.
H317 - May cause an allergic skin reaction.
H318 - Causes serious eye damage.
H330 - Fatal if inhaled.
H334 - May cause allergy or asthma symptoms or breathing difficulties if inhaled.
H335 - May cause respiratory irritation.
H400 - Very toxic to aquatic life.
H411 - Toxic to aquatic life with long lasting effects.

GHS Precaution Phrases: P260 - Do not breathe dust/fume/gas/mist/vapors/spray.
P264 - Wash hands thoroughly after handling.
P270 - Do not eat, drink or smoke when using this product.
P271 - Use only outdoors or in a well-ventilated area.
P272 - Contaminated work clothing should not be allowed out of the workplace.
P273 - Avoid release to the environment.
P280 - Wear protective gloves/protective clothing/eye protection/face protection.
P284 - Wear respiratory protection .
P285 - In case of inadequate ventilation wear respiratory protection.
GHS Response Phrases: P301+310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor/physician.
P303+361+353 - IF ON SKIN (or hair): Remove/take off immediately all contaminated clothing. Rinse skin with water/shower.
P304+340 - IF INHALED: Remove victim to fresh air and keep at rest in a position



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comfortable for breathing.

P305+351+338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P310 - Immediately call a POISON CENTER or doctor/physician.

P330 - Rinse mouth.

P333+313 - If skin irritation or rash occurs, seek medical advice/attention.

P342+311 - If experiencing respiratory symptoms call a POISON CENTER or doctor/physician.

P363 - Wash contaminated clothing before reuse.

P391 - Collect spillage.

GHS Storage and Disposal Phrases:

P403+233 - Store container tightly closed in well-ventilated place.

P405 - Store locked up.

P501 - Dispose of contents/container to an approved waste disposal plant.

Hazard Rating System:

HEALTH	3
FLAMMABILITY	1
PHYSICAL	0
PPE	D

HMIS:

Potential Health Effects (Acute and Chronic):

Chronic exposure may cause effects similar to those of acute exposure.

Inhalation:

May cause respiratory sensitization. May cause respiratory irritation.

Skin Contact:

Causes skin irritation or burns. May cause skin sensitization, an allergic reaction, which becomes evident upon re-exposure to this material.

Eye Contact:

Causes eye irritation or eye damage.

Ingestion:

Harmful if swallowed. May cause gastrointestinal irritation with nausea, vomiting and diarrhea.

3. COMPOSITION/INFORMATION ON INGREDIENTS

CAS #	Ingredient(s)	Concentration
111-30-8	Glutaraldehyde	48.5 -51.5 %

* The remaining balance of the ingredients is water (CAS# 7732-18-5)

4. FIRST AID MEASURES

Emergency and First Aid Procedures:

In Case of Inhalation:

Remove to fresh air and keep at rest in a position comfortable for breathing. If breathing is difficult, give oxygen. Seek medical advice/attention immediately.

In Case of Skin Contact:

Wash exposed area with soap and water thoroughly. Remove contaminated clothing and wash before reuse. Seek medical advice/attention immediately.

In Case of Eye Contact:

Immediately flush eyes with running water for at least 15 minutes. Remove contact lenses if present after the first 5 minutes. Seek medical advice/attention immediately.

In Case of Ingestion:

Rinse mouth with water. Do NOT induce vomiting. Never give anything to an unconscious person. Seek medical advice/attention immediately.

Note to Physician:

Treat symptomatically and supportively.



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5. FIRE FIGHTING MEASURES

Flash Pt: > 200.00 F (93.3 C) Method Used: ASTM D Closed Cup

Explosive Limits: LEL: No data. UEL: No data.

Autoignition Pt: > 395.00 C (743.0 F)

Suitable Extinguishing Media: Use water spray, dry chemical, carbon dioxide, or appropriate foam.

Fire Fighting Instructions: As in any fire, wear a self-contained breathing apparatus in pressure-demand, MSHA/NIOSH (approved or equivalent), and full protective gear. Material will not burn. During a fire, irritating and highly toxic gases may be generated by thermal decomposition or combustion. Use water spray to cool unopened containers.

Flammable Properties and Hazards: No data available.

6. ACCIDENTAL RELEASE MEASURES

Protective Precautions, Protective Equipment and Emergency Procedures: Wear appropriate PPE equipment. See section 8.

Environmental Precautions: Avoid release to the environment.

Steps To Be Taken In Case Material Is Released Or Spilled: Collect liquid by absorbing with an inert material (e.g. vermiculite, dry sand, earth), clean contaminated surface thoroughly, and place in a container for disposal according to local regulations.

7. HANDLING AND STORAGE

Precautions To Be Taken in Handling: Avoid contact with eyes, skin, or clothing. Wash thoroughly after handling. Avoid breathing product vapors or mists. Use with adequate ventilation. Keep containers closed when not in use.

Precautions To Be Taken in Storing: Store in a dry, cool, and well ventilated area away from heat sources and incompatible materials. Keep containers securely sealed. Protect containers from physical damage and check regularly for leaks.

Other Precautions: Segregate from foods and animal feeds.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

CAS #	Components (Chemical Name)	OSHA TWA	ACGIH TWA	Other Limits
111-30-8	Glutaraldehyde	No data.	CEIL: 0.05 ppm	No data.

Respiratory Equipment (Specify Type): Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Eye Protection: Goggles and Face Shield.

Protective Gloves: Wear appropriate protective gloves to prevent skin exposure.

Other Protective Clothing: Wear appropriate protective clothing to prevent skin exposure.

Engineering Controls (Ventilation etc.): Use local and/or general exhaust ventilation to maintain airborne concentrations below irritating levels or airborne exposure limits, whichever is lower. Local exhaust is generally preferred because it can control the emission of the contaminant at its source, preventing dispersion of it into the general work area.

Work/Hygienic/Maintenance Practices: An eye wash station should be accessible in the immediate area of use. Always observe good personal hygiene measures, such as washing after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing and protective equipment



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to remove contaminants. Discard contaminated clothing and footwear that cannot be cleaned. Practice good housekeeping.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical States: ☐ Gas ☒ Liquid ☐ Solid
Appearance and Odor: colorless to yellow.
characteristic odor.
Freeze Point/Pour Point: -21.00 C (-5.8 F)
Boiling Point: > 100.00 C (212.0 F)
Decomposition Temperature: 188.00 C (370.4 F)
Autoignition Pt: > 395.00 C (743.0 F)
Flash Pt: > 200.00 F (93.3 C) Method Used: ASTM D Closed Cup
Explosive Limits: LEL: No data. UEL: No data.
Specific Gravity (Water = 1): 1.13+/-0.05
Vapor Pressure (vs. Air or mm Hg): 0.3 MM_HG at 20.0 C (68.0 F)
Vapor Density (vs. Air = 1): No data.
Evaporation Rate: No data.
Solubility in Water: soluble
pH: 3.0 - 5.0
Percent Volatile: No data.

10. STABILITY AND REACTIVITY

Stability: Unstable ☐ Stable ☒
Conditions To Avoid - Instability: Incompatible materials.
Incompatibility - Materials To Avoid: Acids. Bases, Amines, Alcohols.
Hazardous Decomposition or Byproducts: Carbon oxides.
Will occur ☐ Will not occur ☒
Possibility of Hazardous Reactions: Polymerizes slowly on exposure temperature rise. May polymerize on exposure to some compounds e.g.: amines
Conditions To Avoid - Hazardous Reactions:



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11. TOXICOLOGICAL INFORMATION

Toxicological Information:	The product has not been tested. Any statements on toxicology have been derived from products of a similar structure and composition.
Carcinogenicity/Other Information:	CAS# 111-30-8: Acute toxicity, LD50, Oral, Rat, 134.0 MG/KG. Result: Effects on Embryo or Fetus: Fetotoxicity (except death, e.g., stunted fetus). Specific Developmental Abnormalities: Musculoskeletal system. - Oyo Yakuri. Pharmacometrics., Oyo Yakuri Kenkyukai, CPO Box 180, Sendai 980-91 Japan, Vol/p/yr: 19,503, 1980 Acute toxicity, LC50, Inhalation, Rat, 480.0 MG/M3, 4 H. Result: Behavioral: Somnolence (general depressed activity). Behavioral: Coma. Lungs, Thorax, or Respiration: Respiratory stimulation. - United States Environmental Protection Agency, Office of Pesticides and Toxic Substances., U.S. Environmental Protection Agency, Office of Pesticides and Toxic Substances, 401 M St., SW, Washington, DC 20460, Vol/p/yr: 8EHQ-1290-, Acute toxicity, LD50, Skin, Rat, > 2500. MG/KG. Result: Skin and Appendages: Skin: After systemic exposure: Dermatitis, other. - Oyo Yakuri. Pharmacometrics., Oyo Yakuri Kenkyukai, CPO Box 180, Sendai 980-91 Japan, Vol/p/yr: 12,11, 1976 CAS# 111-30-8: Not listed by ACGIH, IARC, NTP, or CA Prop 65.

CAS #	Ingredient(s)	NTP	IARC	ACGIH	OSHA
111-30-8	Glutaraldehyde	n.a.	n.a.	A4	n.a.

12. ECOLOGICAL INFORMATION

General Ecological Information:	The product has not been tested. Any statements on eco-toxicology have been derived from products of a similar structure and composition.
Results of PBT and vPvB assessment:	CAS# 111-30-8: LC50, Bluegill (<i>Lepomis macrochirus</i>), 22.60 PPM, 96 H, Mortality. Result: Age Effects. - Pesticide Ecotoxicity Database (Formerly: Environmental Effects Database (EEDB)), Office of Pesticide Programs, 2000 Effective concentration to 50% of test organisms., Water Flea (<i>Daphnia magna</i>), 0.750 PPM, 48 H, Intoxication., Result: Abnormal development. - Pesticide Ecotoxicity Database (Formerly: Environmental Effects Database (EEDB)), Office of Pesticide Programs, 2000 Effective concentration to 50% of test organisms., Green Algae (<i>Pseudokirchneriella</i>



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subcapitata), Exponential Growth Phase, 3.950 MG/L, 48 H, Physiology, Water
temperature: 24.00 C (75.2 F) C.

Result:

Age Effects.

- Individual and Combined Toxicity of Nitriles and Aldehydes to Raphidocelis subcapitata,
Chen, C.Y., S.L. Chen, and E.R. Christensen, 2005

13. DISPOSAL CONSIDERATIONS

Waste Disposal Method:

RCRA: It is the responsibility of the product user to determine at the time of disposal whether a material containing the product or derived from the product should be classified as a hazardous waste (40 CFR 261.20-24).

Disposal: Whatever cannot be saved for recovery or recycling should be disposed of in accordance with local, state, and federal regulations. Consult state or local regulation for any additional handling, treatment, or disposal requirements. For disposal, contact a properly licensed waste treatment, storage, disposal, or recycling facility.

14. TRANSPORT INFORMATION

LAND TRANSPORT (US DOT):

DOT Proper Shipping Name: Corrosive liquids, toxic, n.o.s.(Glutaraldehyde)

DOT Hazard Class: 8 (6.1) CORROSIVE, POISON

UN/NA Number: UN2922

Packing Group:

II



LAND TRANSPORT (Canadian TDG):

TDG Shipping Name: Corrosive liquids, toxic, n.o.s.(Glutaraldehyde)

UN Number: 2922

Packing Group:

II

Hazard Class: 8 (6.1) - CORROSIVE,
POISON

TDG Classification:

MARINE TRANSPORT (IMDG/IMO):

IMDG/IMO Shipping Name: Corrosive liquids, toxic, n.o.s.(Glutaraldehyde)

UN Number: 2922

Packing Group:

II

Hazard Class: 8 (6.1) - CORROSIVE,
POISON

IMDG MFAG Number:

Marine Pollutant:

Yes

IMDG EMS Page:

AIR TRANSPORT (ICAO/IATA):

ICAO/IATA Shipping Name: Corrosive liquids, toxic, n.o.s.(Glutaraldehyde)

UN Number: 2922

Packing Group:

II

Hazard Class: 8 (6.1) - CORROSIVE,
POISON



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15. REGULATORY INFORMATION

EPA SARA (Superfund Amendments and Reauthorization Act of 1986) Lists

CAS #	Ingredient(s)	S. 302 (EHS)	S. 304 RQ	S. 313 (TRI)
111-30-8	Glutaraldehyde	No	No	No

This material meets the EPA 'Hazard Categories' defined for SARA Title III Sections 311/312 as indicated:

☒ Yes	☐ No	Acute (immediate) Health Hazard
☒ Yes	☐ No	Chronic (delayed) Health Hazard
☐ Yes	☒ No	Fire Hazard
☐ Yes	☒ No	Sudden Release of Pressure Hazard
☐ Yes	☒ No	Reactive Hazard

CAS #	Ingredient(s)	Canadian NPRI	Canadian IDL	Canadian DSL
111-30-8	Glutaraldehyde	No	Yes	Yes

CAS #	Ingredient(s)	Other US EPA or State Lists
111-30-8	Glutaraldehyde	CAA HAP, ODC: No; CWA NPDES: No; TSCA: Yes - Inventory, 8A PAIR; CA PROP.65: No

CAS #	Ingredient(s)	International Regulatory Lists
111-30-8	Glutaraldehyde	

Canadian WHMIS Classification:

No data available.

Regulatory Information: Primary EPA Registration # : 88714-10
This product is subject to registration under FIFRA.
All ingredients are reported in the EPA TSCA Inventory.

Canadian regulations: This product is not registered for sale or use as a pesticide in Canada.

16. OTHER INFORMATION

Revision Date: 12/05/2017

Additional Information About This Product: No data available.

Company Policy or Disclaimer:

The information accumulated herein is believed to be accurate based on the information provided, although no guarantee or warranty, either expressed or implied is made as to the accuracy or completeness of this information, whether originating with this company or not. Recipients are advised to confirm in advance of need that the information is correct, applicable and suitable to their circumstances. The conditions or methods of handling, storage, use and disposal of the product and container are beyond our control and may be beyond our knowledge. For this and other reasons, we do not assume responsibility and expressly disclaim liability for loss, damage or expense arising out of or in any way connected with the handling, storage or use of this information or product. If the product is used as a component in another product, this information may not be applicable.